Literature Review

Project: YojanaBench — Open Benchmark for India Welfare-Scheme Document Reasoning, Multi-Agent Audit Pipeline and DBT Money-Trace Grievance Module

This review surveys the published work related to this project, grouped into four themes, identifies the **research gaps** each work leaves open, and maps those gaps to what this project contributes. Every reference listed here was checked against its original source (paper page, journal site, arXiv or Government of India press release).

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1. Scope and Method of the Survey

The literature was collected under four search themes:

Theme	Question being asked	Papers
A. AI benchmarks	Do existing benchmarks test *this kind* of task?	GAIA, DocVQA, LawBench, LegalBench
B. Core methods	What techniques does the project build upon?	RAG survey, LLM multi-agent survey
C. AI for welfare schemes	What already exists for Indian scheme access?	Vaishnav & Nathan, myScheme study, hybrid-RAG chatbots, AgriGov, PM-KISAN chatbot
D. DBT, leakage & fraud	What exists for tracing money and catching fraud?	DBT impact studies, AI-for-DBT-integrity, RBI MuleHunter.AI, mule-detection research

Inclusion criteria: papers/directly relevant systems that either (1) define a benchmark or evaluation method, (2) provide a technique used by this project, or (3) solve a part of the same real-world problem (scheme discovery, eligibility, DBT leakage).

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2. Theme A — General-Purpose and Domain AI Benchmarks

2.1 GAIA: A Benchmark for General AI Assistants

Mialon et al., 2023 [1]

GAIA is a benchmark of 466 real-world questions testing reasoning, multimodality and tool use in general-purpose assistants. It is an important reference point because it shows how a *benchmark* should be built: real questions, exact ground-truth answers, and reproducible scoring.

Relevance to this project: GAIA demonstrated that honest evaluation needs an answer key that a human verified — the same principle YojanaBench applies, but to a completely different task class.

Limitation: GAIA is general-purpose and English-only. It has no domain documents, no eligibility or entitlement reasoning, no verdict ground truth (`eligible / ineligible / needs_info`), and nothing India-specific. A welfare scheme cannot be evaluated by a general assistant benchmark.

2.2 DocVQA: A Dataset for VQA on Document Images

Mathew, Karatzas, Jawahar, 2021 [2]

DocVQA contains 50,000 extractive question-answering pairs over 12,000+ document images. It is the standard reference for "questions answered from documents".

Relevance: YojanaBench's QA pairs over scheme guideline documents belong to the same *document-QA* family, so DocVQA defines the evaluation vocabulary (exact match, F1) that YojanaBench reuses.

Limitation: DocVQA answers are **span extractions** — copy the text from the page. It has no eligibility verdicts, no numerical entitlement computation, no counterfactual (boundary) scenarios, and no clause-level evidence labels. Knowing *what the document says* is not the same as deciding *whether a person qualifies*.

2.3 LawBench: Benchmarking Legal Knowledge of LLMs

Fei et al., 2024 [3]

LawBench evaluates 20 Chinese legal tasks across three cognitive levels (memorisation, understanding, application) — one of the first domain-specific legal benchmarks for LLMs.

Relevance: it provides a template for organising a domain benchmark by *cognitive difficulty*, which YojanaBench follows with its base-case vs edge-case (counterfactual) split.

Limitation: it is confined to Chinese law. It contains no welfare-scheme documents, no benefit-amount computation, no counterfactual eligibility scenarios, and evaluates single-model prompting rather than a multi-agent audit pipeline.

2.4 LegalBench: A Collaboratively Built Benchmark for Legal Reasoning

Guha et al., 2024 [4]

LegalBench is a collaboratively built set of 162 tasks measuring US legal reasoning (issue spotting, rule application, reasoning over contracts and case law).

Relevance: it shows how a large community can build many small, focused tasks with clear labels — the model YojanaBench uses for its QA categories (eligibility / entitlement / documents / appeal).

Limitation: it is US case law and contracts. It does not address government benefit rules, document-compliance checklists, or numeric entitlement calculation — the everyday "bureaucratic" law that citizens actually face.

Theme A conclusion: benchmarks exist for *general questions*, *document extraction*, and *legal reasoning* — but none of them measures **scheme-eligibility decisions with computed benefits and tri-state verdicts** on Indian welfare documents. That is the first gap.

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3. Theme B — Core Techniques: RAG and Multi-Agent LLM Systems

3.1 Retrieval-Augmented Generation for Large Language Models: A Survey

Gao et al., 2023 [5]

This survey organises RAG into *naive*, *advanced* and *modular* paradigms and reviews how retrieved evidence improves factuality. It is the standard citation for why an LLM should read the rulebook before answering.

Relevance: YojanaBench's RAG agent follows the survey's guidance — retrieve first, then generate — and additionally combines **lexical (BM25) and semantic (TF-IDF) retrieval** in the "hybrid" style the survey describes. Clause IDs returned with every snippet give the provenance the survey calls for.

Limitation: the survey offers no *domain benchmark* for governance documents, and its evaluation ideas (answer similarity) lack the **verdict-level** and **computation-level** metrics needed for high-stakes compliance decisions — you cannot score an eligibility decision with string similarity alone.

3.2 Large Language Model based Multi-Agents: A Survey of Progress and

Challenges
Guo et al., 2024 [6]

The survey systematically reviews LLM multi-agent designs: how agents are profiled, how they communicate, and how capabilities are acquired.

Relevance: it justifies the architecture used here — specialised agents with single responsibilities and an orchestrator that merges their outputs. YojanaBench's 11-agent audit pipeline and 7-agent money-trace pipeline are a concrete instance of the patterns this survey describes.

Limitation: the survey lists **no welfare-governance application** and no verified benchmark on which an audit-style agent pipeline can be scored against ground truth. Multi-agent systems are proposed faster than they are *measured*.

Theme B conclusion: the *methods* (RAG, multi-agent orchestration) are mature; the *evaluation of those methods on governance tasks* is missing. That is the second gap.

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4. Theme C — Existing Work on Indian Welfare-Scheme Access

4.1 Need of RAG for Conversational Assistants for Government Social Welfare

Scheme Popularization
Vaishnav & Nathan, 2024 [7]

The authors built a RAG chatbot over 2,980 scraped government schemes and measured response similarity, uncertainty and hallucination. They show that retrieval-augmented generation **substantially reduces hallucination** compared with a bare LLM, and that larger models perform better.

Relevance: this is the closest prior work to the retrieval side of YojanaBench, and its findings directly motivate grounding every answer in official scheme text.

Limitation: it has **no human-verified QA ground truth**, no eligibility verdict labels, no entitlement computation checks, and no counterfactual or edge-case evaluation. It measures whether answers *sound* similar to expected responses — not whether the decision is *right*.

4.2 First Impressions from Comparing Form-Based and Conversational

Interfaces for Public Service Access in India
Chaitra C R et al., 2025 [8]

A usability study comparing the official **myScheme** form application with a RAG conversational prototype ("Prabodhini") on 15 users, reporting which interface people prefer and how fast each is.

Relevance: it documents the real user-facing alternative to this project — the myScheme portal — and confirms that citizens find conversational input easier.

Limitation: it evaluates **usability**, not **correctness**. Nothing in the study checks whether the scheme information returned matches the official guideline documents. A fast but wrong answer still scores well in a usability test. Interface studies leave open the question: *is the answer true?*

4.3 AI-Driven Chatbot for Government Scheme Accessibility in India Using

Hybrid RAG with LLMs
Vaishnavi P C & Paramarthalingam, 2024 [9]

Combines BM25 lexical retrieval with dense vector embeddings, adds cross-encoder re-ranking and a rule-based personalisation layer (age, gender, state) over ~3,500 schemes / ~70,000 text segments, using a locally deployed

model for controlled generation.

Relevance: independent confirmation that **hybrid retrieval (BM25 + semantic)** is the right choice for scheme documents — the same combination YojanaBench implements in pure Python. Its local-model requirement also matches this project's offline, zero-cost design.

Limitation: the personalisation layer is a *filter*, not an *auditor*: there is no tri-state verdict, no independent cross-check, no computed entitlement value scored against ground truth, and no benchmark on which the system's decisions can be reproduced and compared.

4.4 AI-Driven Welfare Scheme Recommendation Using Random Forest and RAG

Karunamurthy & Barath, 2026 [10]

A multi-output Random Forest predicts eligibility probabilities across all schemes; a calibrated threshold selects candidates; a RAG module then generates explanations restricted to retrieved verified chunks — so the LLM "cannot recommend a scheme the classifier did not flag eligible".

Relevance: the strongest prior art for **separating the decision from the language**: a deterministic model decides, the LLM only explains. YojanaBench applies the same principle — its deterministic rule engine decides, and the optional LLM only polishes wording.

Limitation: evaluation is on **synthetic datasets** mimicking real records, with no human-verified answer key, no tri-state "needs information" outcome, no counterfactual advice, and no public leaderboard for third-party comparison.

4.5 AgriGov: A Structured Multilingual Dataset Curation for Indian

Government Schemes for Farmers
Bilal & Gopakumar, 2026 [11]

A trilingual (English–Hindi–Marathi) corpus of 50 government schemes scraped from trusted portals, structured into semantic fields (title, eligibility, application process, documents, exclusions), with ~2,100 human-corrected translated segments growing to ~8,000 aligned pairs.

Relevance: demonstrates the value of **structured scheme schemas** — the same idea as YojanaBench's `data/schemes/*.json`, where eligibility clauses, entitlement formulas and document lists are explicit fields rather than free text.

Limitation: AgriGov is a *dataset for translation, QA and retrieval*; it carries no verdict ground truth, no entitlement formulas, no edge cases, and no evaluation pipeline that scores decisions.

4.6 Government deployment: the PM-KISAN AI chatbot

Government of India, Ministry of Agriculture & Farmers Welfare [12]

The Government has launched an official AI chatbot integrated with the PM-KISAN flagship scheme to answer beneficiary queries.

Relevance: shows the problem is recognised at the highest level — scheme Q&A is now a real government service. It also raises the stakes: public-facing scheme AI **must** be evaluated, because a wrong eligibility answer harms a citizen.

Limitation: like the chatbots above, it answers questions; it publishes no benchmark, no accuracy figures against an answer key, and no audit trail of which rule produced which answer.

Theme C conclusion: existing systems focus on **finding** schemes and **conversing** about them. None of them provides an open, reproducible evaluation of whether the *decision* is correct, whether the *money* is right, or what to do when the citizen fails. That is the third gap.

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5. Theme D — DBT, Leakage and Mule-Account Fraud

5.1 Impact Assessment of the Direct Benefit Transfer Scheme in India: A

Case Study of Haryana
IJCRT, 2023 [13]

Reports that DBT (the Jan-Dhan–Aadhaar–Mobile "JAM triangle") reduced intermediaries: 85% of rural and 69% of urban households benefited; Haryana's identity-linked transfers saved ₹1,200 crore by detecting fake beneficiaries; pre-DBT, historically only about 15 paise of every rupee reached the intended recipient.

Relevance: establishes the *economic importance* of the money-trace problem this project attacks — leakage is large, measurable, and still partially unsolved.

Limitation: impact studies work at the *aggregate* level (how much leakage overall). They do not answer the individual citizen's question: *"the record says paid — where did my money go?"*

5.2 Leveraging AI and ML to Enhance the Integrity of Social Protection and

DBT Systems
IJCSP, 2025 [14]

Applies classification and anomaly-detection models to DBT/MGNREGS transaction data, reporting up to ~27% reduction in false beneficiary inclusion and ~91% accuracy in forecasting fraudulent transaction patterns.

Relevance: confirms that machine learning can find DBT fraud at scale, and motivates including a fraud-detection stage in a grievance pipeline.

Limitation: statistical fraud models need large labelled training sets and produce *probabilities*, not *explainable case verdicts*. A beneficiary (or an officer) needs a concrete statement — "₹500 short, credited to account ending 4821" — with evidence, not an anomaly score.

5.3 RBI's MuleHunter.AI

Reserve Bank of India / Press Information Bureau [15]

The RBI has rolled out **MuleHunter.AI**, an AI/ML solution specifically for **mule account detection** — accounts used to divert and layer funds in cyber frauds — alongside revised instructions on unauthorised transactions and expanded AI-driven fraud analytics.

Relevance: validates this project's design choice to treat *shared receiving accounts* as a first-class fraud signal. YojanaBench's rule "an account receiving benefits for ≥ 3 different beneficiaries is a mule ring" is a transparent, explainable cousin of exactly what the RBI is automating.

Limitation: MuleHunter.AI is a bank-side monitoring system. It does not connect a **welfare complaint** to the **sanction order** and the **credit record**, nor does it explain to the citizen what happened to their money.

5.4 Money-Mule Detection as a Relational Learning Problem: A Study on the

RBIH NFPC Dataset
Ahmad & Deodhar, 2026 [16]

Shows that a mule's individual transfers look ordinary — what reveals the account is its *relationships* (shared counterparties, addresses) and its *temporal shape* (burst in, burst out, silence). The paper argues that production systems remain largely rule-based and that research usually classifies transactions instead of accounts.

Relevance: provides the theoretical backing for account-level reasoning in the fraud agent: YojanaBench looks at the *account's* history across beneficiaries rather than judging one payment in isolation.

Limitation: it is a machine-learning study on internal banking data — not a public, reproducible system, and not aimed at grievance redressal for welfare beneficiaries.

Theme D conclusion: fraud research is bank-centred and aggregate; grievance systems are form-centred and stop at "complaint registered". Nothing joins them into an **auditable, citizen-facing money trace** with an explainable verdict and a routed escalation notice. That is the fourth gap.

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6. Comparative Summary of Related Work

Work	Year	Task	Verdict ground truth	Entitlement maths	Counterfactual / edge cases	Human-verified QA	Public leaderboard	Post-disbursement trace
GAIA [1]	2023	General QA	—	—	—	✓	—	—
DocVQA [2]	2021	Document span QA	—	—	—	✓	—	—
LawBench [3]	2024	Chinese legal tasks	—	—	—	✓	—	—
LegalBench [4]	2024	US legal reasoning	—	—	—	✓	—	—
RAG Survey [5]	2023	Method survey	—	—	—	—	—	—
Multi-Agent Survey [6]	2024	Method survey	—	—	—	—	—	—
Vaishnav & Nathan [7]	2024	Scheme chatbot	—	—	—	—	—	—
myScheme interface study [8]	2025	Usability	—	—	—	—	—	—
Hybrid-RAG chatbot [9]	2024	Scheme retrieval	—	—	—	—	—	—
Random Forest + RAG [10]	2026	Recommendation	—	—	—	—	—	—
AgriGov [11]	2026	Dataset	—	—	—	—	—	—
DBT Haryana study [13]	2023	Impact	—	—	—	—	—	aggregate only
AI/ML for DBT integrity [14]	2025	Fraud ML	—	—	—	—	—	aggregate only
Money-mule relational ML [16]	2026	Fraud ML	—	—	—	—	—	account-level
YojanaBench (this work)	**2026**	**Scheme audit + money trace**	**✓ tri-state**	**✓ deterministic**	**✓**	**✓**	**✓**	**✓ 7-agent trace**

(✓ = provided by the work; — = not addressed)

7. Research Gaps Identified

Gap	Statement of the gap	Where it was observed
G1 — No domain benchmark	No benchmark scores Indian welfare-scheme reasoning with verdicts, entitlement values and cited clauses	[1][2][3][4] cover general, document, legal — not schemes
G2 — Extraction ≠ decision	Document QA extracts text; it never decides *eligibility* or handles *missing information*	[2] spans only; [7][9] answers only
G3 — Evaluation without truth	Scheme chatbots are evaluated by similarity/usability, not against a human-verified answer key	[7][8][9][10]
G4 — No honesty about uncertainty	Systems are forced-choice (Yes/No) or free-text; none emits a first-class **"needs information"** verdict with the missing fields named	[7][9][10][12]
G5 — Decision/maths not separated or verified	Where arithmetic appears (benefit amounts), it is produced by the language model or not computed at all; no independent cross-check agent	[7][9][10]
G6 — No post-disbursement accountability	Fraud/DBT work is aggregate or bank-internal; no public system traces one complaint from sanction → credit with an explainable verdict and escalation notice	[13][14][15][16]
G7 — No reproducible comparison	No one can score a third-party model on the same scheme items — there is no shared answer key or leaderboard	all of Theme C

8. How This Project Addresses Each Gap

Gap	YojanaBench's answer	Where in this project
G1	35 schemes, human-verified QA pairs, verdict ground truth, entitlement targets	`data/schemes/*.json`, `data/qa/qa_benchmark.json`
G2	Every QA item carries a decision + computed value + cited clause IDs, not a text span	`entitlement_agent.py`, `orchestrator.py`
G3	Scoring with EM, token-F1, verdict accuracy, entitlement error, document F1, ECE — all against the answer key	`evaluation_agent.py`
G4	**Tri-state rule engine** (`True / False / Unknown`) → `eligible / ineligible / needs_info` , with the missing fields explicitly listed	`condition_eval.py`
G5	Benefits computed in deterministic Python (never by an LLM) + an **independent Verification agent**; disagreement forces human review; ablation quantifies both	`entitlement_agent.py`, `verification_agent.py`, `ablation.py`
G6	**Seven-agent DBT money trace**: sanction → ledger → credit check → reconciliation → mule-ring fraud → verdict → routed notice	`backend/trace/*`
G7	Open leaderboard: anyone can score their model on the same items with the same metrics	`benchmark.py`, `leaderboard.py`

Beyond the gaps: two advisory features that no surveyed work provides — the **Counterfactual Advisor** ("the smallest change that would make you eligible", verified by re-running the rules) and the **Document-Gap Analyzer**

("will submitting these papers even change the verdict?").

9. Conclusion

The literature shows a clear progression: from general QA benchmarks (GAIA, DocVQA) to domain legal benchmarks (LawBench, LegalBench), from naive LLM answers to grounded RAG, and from single models to multi-agent systems. On the Indian welfare side, tools such as myScheme and several recent RAG chatbots solve **discovery and conversation**, while DBT research solves **aggregate leakage and bank-side fraud detection**.

The intersection — **an open, auditable evaluation of scheme decisions, and a citizen-facing trace of where a specific payment went** — is empty. That intersection is precisely where YojanaBench sits. Its claims are deliberately narrow and defensible: not "the first scheme finder", but, to our knowledge, one of the first **open, reproducible benchmarks** for welfare-scheme document reasoning — with tri-state verdicts, clause-cited audits, deterministic benefit maths, verified counterfactual advice, and a seven-agent DBT money trace.

10. References

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[16] O. Ahmad and A. Deodhar, "Money-Mule Detection as a Relational Learning Problem: A Study on the RBIH NFPC Dataset," arXiv:2610.21314, 2026.

Note: references [1]–[8] are carried over from the project's original literature survey table (`YojanaBench_Literature_Survey.pdf`); [9]–[16] were added for this review and verified against the publisher's or arXiv's record.